

Intermediate Level

Introduction to
Computing at CARC
1 hour version with CASA

2nd in a 3 part series: 1) BASH, 2) SLURM, then
3) Parallelism

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Version 0.1



Goals

- 1) SLURM scheduler literacy
- 2) Running CASA interactively and in batch
- 2) JupyterHub and CASA
- We wont cover file transfer, storage systems, module system, conda, PBS. (These are all covered in depth in the video tutorials)

Logging into Hopper



First login to the Linux **workstation** in front of you. Your CARC username is on the sign in sheet.

If you have logged in before, use your **existing password**

Otherwise, your initial password is **Welcome2CARC**

This is an “important step” so don’t let me move on until you have logged in

Logging into Hopper



```
ssh vanilla@hopper.alliance.unm.edu
```

Should prompt you for a password...

Don't let me move on until you are able to login.

Replace vanilla with your name (unless your last name is Ice)

Logging into Hopper

Welcome to Hopper

Be sure to review the "Acceptable Use" guidelines posted on the CARC website.

For assistance using this system email help@carc.unm.edu.

Tutorial videos can be accessed through the CARC website: Go to <http://carc.unm.edu>, select the "New Users" menu and then click "Introduction to Computing at CARC".

Warning: By default home directories are world readable. Use the `chmod` command to restrict access.

Don't forget to acknowledge CARC in publications, dissertations, theses and presentations that use CARC computational resources:

"We would like to thank the UNM Center for Advanced Research Computing, supported in part by the National Science Foundation, for providing the research computing resources used in this work."

Please send citations to publications@carc.unm.edu.

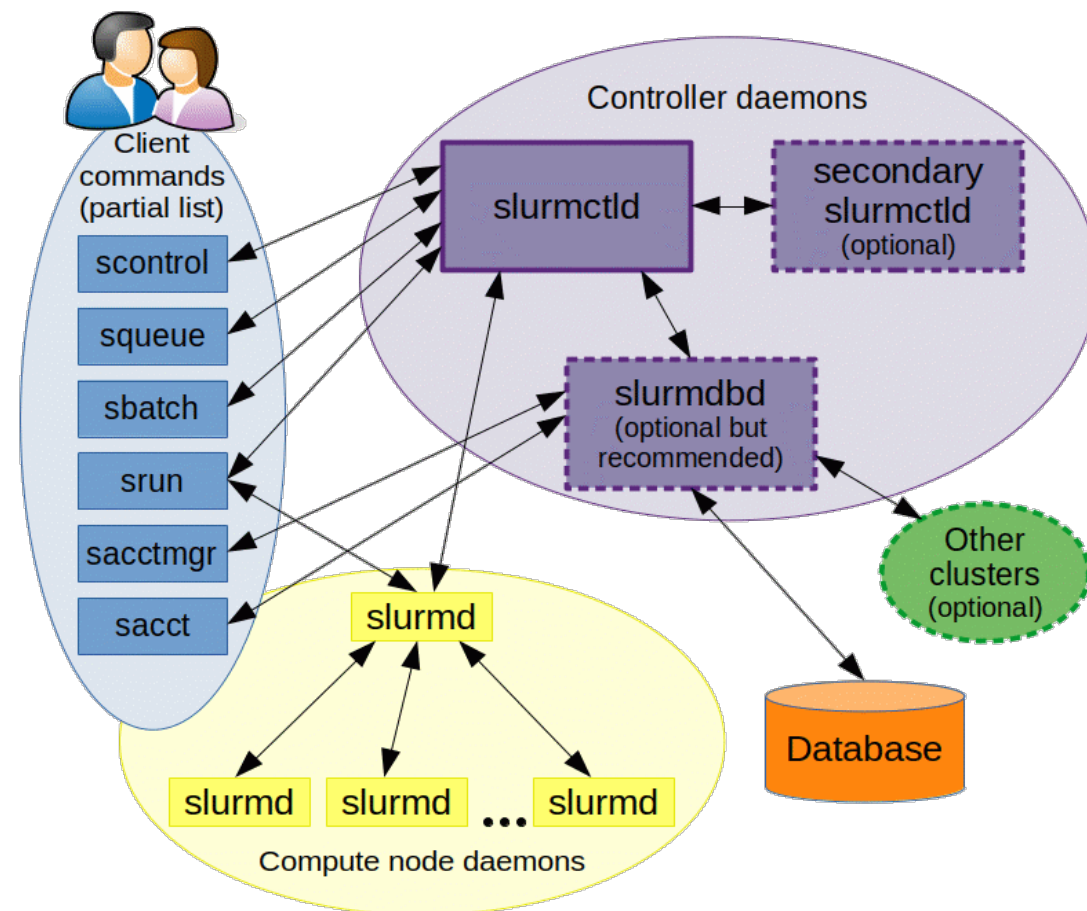
There are three types of slurm partitions on Hopper:

- 1) General - this partition is accessible by all CARC users.
- 2) Condo - preemptable scavenger queue available to all condo users. Your job must use checkpointing to use this queue or you will lose any work you have done if it is preempted by the partition's owner.
- 3) Named partitions - these partitions are available to condo users working under the grant/lab/center that purchased the associated hardware.

Type "qgrok" to get the status of the partitions.

```
-----  
Last login: Wed Jul 27 17:46:13 2022 from 129.24.246.68  
vanilla@hopper:~ $
```

Simple Linux Utility for Resource Management





VOTED **#1** SOFT DRINK OF THE 31ST CENTURY!



SLURP'S ADVERTISING

[vanilla@hopper ~]\$ qgrok

queues	free	busy	offline	jobs	nodes	CPUs	GPUs	CPUs/node	GPUs/node	Memory/node	time_limit	CPU_limit
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
general	4	6	0	4	10	320	0	32	0	93G	2-00:00:00	64
debug	2	0	0	0	2	64	0	32	0	93G	4:00:00	8
condo	22	25	4	7	51	1632	28	32	2	93G-1.5T	2-00:00:00	192
bugs	2	0	0	0	2	64	0	32	0	93G	7-00:00:00	

```
vanilla@hopper:~ $ qgrok
```

queues	free	busy	offline	jobs	nodes	CPUs	GPUs
-----	-----	-----	-----	-----	-----	-----	-----
general	1	9	0	7	10	320	0
debug	2	0	0	0	2	64	0
condo	18	19	1	7	38	1216	8
bugs	0	2	0	0	2	64	0
pcnc	1	1					
pathogen	1	0					
tc	5	5					
gold	2	0					
fishgen	0	1					
neuro-hsc	8	6					
cup-ecs	0	2					
tid	0	1					
biocomp	0	1					
chakra	1	0					
pna	0	0					
totals:	19	28					

**Open partitions for use by
everyone with a CARC account.**

**Purchased by the Office for the
Vice President for Research.**

vanilla@hopper:~ \$ qgrok

queues	free	busy	offline	jobs	nodes	CPUs	GPUs
-----	-----	-----	-----	-----	-----	-----	-----
general	1	9	0	7	10	320	0
debug	2	0	0	0	2	64	0
condo	18	19	1	7	38	1216	8
bugs	0	2	0	0	2	64	0
pcnc	1	1	0	0	2	64	0
pathogen	1	0	0	0	1	32	0
tc	5	5	0	3	10	320	0
gold	2	0	0	0	2	64	0
fishgen	0	1	0	0	1	32	0
neuro-hsc	8	6	0	0	14	448	0
cup-ecs	0	2	0	2	2	64	4
tid	0	1	0	0	1	32	2
biocomp	0	1	0	0	1	32	1
chakra	1	0	0	0	1	32	1
pna	0	0	1	0	1	32	0
totals:	19	28	1	14	48	1536	8

```
vanilla@hopper:~ $ qgrok
```

queues	free	busy	offline	jobs	nodes	CPUs	GPUs
-----	-----	-----	-----	-----	-----	-----	-----
general	1	9	0	7	10	320	0
debug	2	0	0	0	2	64	0
condo	18	19	1	7	38	1216	8
bugs	0	2	0	0	2	64	0
pcnc	1	1					
pathogen	1	0					
tc	5	5					
gold	2	0					
fishgen	0	1					
neuro-hsc	8	6					
cup-ecs	0	2					
tid	0	1					
biocomp	0	1					
chakra	1	0					
pna	0	0					
totals:	19	28					

Private partitions

- Reserved for use by the purchaser.
- Request access by emailing support@carc.unm.edu and CC the partition owner.

```
vanilla@hopper:~ $ qgrok
```

queues	free	busy	offline	jobs	nodes	CPUs	GPUs
-----	-----	-----	-----	-----	-----	-----	-----
general	1	9	0	7	10	320	0
debug	2	0	0	0	2	64	0
condo	18	19	1	7	38	1216	8
bugs	0	2	0	0	2	64	0
pcnc	1	1					
pathogen	1	0					
tc	5	5					
gold	2	0					
fishgen	0	1					
neuro-hsc	8	6					
cup-ecs	0	2					
tid	0	1					
biocomp	0	1					
chakra	1	0					
pna	0	0					
totals:	19	28					

Condo “scavenger” partition

- Allows you to use compute nodes purchased by another group that are currently idle.
- May be interrupted at any time if the owners start to use it.

```
[vanilla@hopper ~]$ quotas
```

```
Home Directory (/users/vanilla):
```

```
quota: Cannot resolve mountpoint path /root/.spack: Permission denied
```

```
Disk quotas for user vanilla (uid 659):
```

Filesystem	space	quota	limit	grace	files	quota	limit	grace
chama:/home/homes	1527M	100G	200G		14913	4295m	4295m	

```
Centerwide user scratch (/carc/scratch/users/vanilla)
```

```
Quota information for storage pool Default (ID: 1):
```

user/group			size			chunk files	
name	id		used	hard		used	hard
vanilla	1512		592.71 GiB	1024.00 GiB		32784	unlimited

```
Centerwide scratch quota for project vanilla2016174 (/carc/scratch/projects/vanilla2016174)
```

```
Quota information for storage pool Default (ID: 1):
```

user/group			size			chunk files	
name	id		used	hard		used	hard
vanilla2016174	2016142		190.97 GiB	1024.00 GiB		23704	unlimited

```
[vanilla@hopper ~]$ sinfo --partition debug
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
debug      up        4:00:00    2    idle hopper[011-012]
```

sinfo reports information about
partitions

```
[vanilla@hopper ~]$ sinfo --partition debug
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
debug      up       4:00:00    2    idle hopper[011-012]
```

The debug queues are intended
for testing your programs.

And for interactive jobs.

```
[vanilla@hopper ~]$ sinfo --partition debug
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
debug      up        4:00:00    2    idle hopper[011-012]
```



Name

```
[vanilla@hopper ~]$ sinfo --partition debug
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
debug      up       4:00:00    2    idle hopper[011-012]
```



You can run a “job” for up to 4 hrs.

```
[vanilla@hopper ~]$ sinfo --partition debug
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
debug      up        4:00:00    2    idle hopper[011-012]
```



There are two nodes in this partition.

```
[vanilla@hopper ~]$ sinfo --partition debug
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
debug      up        4:00:00    2   idle hopper[011-012]
```



The names of the nodes in the
partition

```
[vanilla@hopper ~]$ sinfo --partition debug
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
debug      up        4:00:00    2   idle hopper[011-012]
```



The names of the nodes in the
partition

```
[vanilla@hopper ~]$ sinfo --partition general
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
general*   up    2-00:00:00      9  alloc hopper[001-009]
general*   up    2-00:00:00      1  idle  hopper010
```



Hopper001 through 009 are running jobs.

Hopper010 is waiting to be used.

```
[vanilla@hopper ~]$ hostname  
hopper  
[vanilla@hopper ~]$
```



Running on the Head Node.

The head node's name is "hopper".

```
[vanilla@hopper ~]$ hostname  
hopper
```

```
[vanilla@hopper ~]$ man hostname
```

```
[vanilla@hopper ~]$ hostname  
hopper
```

```
[vanilla@hopper ~]$ man hostname  
(‘q’ to quit)
```

```
[vanilla@hopper ~]$ man man  
(‘q’ to quit)
```

```
[vanilla@hopper ~]$ man sinfo
```

```
sinfo(1)  
Commands
```

```
Slurm  
sinfo(1)
```

NAME

sinfo - View information about Slurm nodes and partitions.

SYNOPSIS

sinfo [OPTIONS...]

DESCRIPTION

sinfo is used to view partition and node information for a system running Slurm

OPTIONS

-a, --all

Display information about all partitions. This causes information to be displayed about partitions that are configured as hidden and partitions that are unavailable to the user's group.

```
[vanilla@hopper ~]$ sinfo --all
```

PARTITION	AVAIL	TIMELIMIT	NODES	STATE	NODELIST
general*	up	2-00:00:00	9	alloc	hopper[001-009]
general*	up	2-00:00:00	1	idle	hopper010
debug	up	4:00:00	2	idle	hopper[011-012]
condo	up	2-00:00:00	1	down*	hopper045
condo	up	2-00:00:00	3	mix	hopper[018-020]
condo	up	2-00:00:00	16	alloc	hopper[013-015,028-036,049-052]
condo	up	2-00:00:00	18	idle	hopper[016-017,021-027,037-044,053]
bugs	up	7-00:00:00	2	alloc	hopper[013-014]
pcnc	up	7-00:00:00	1	alloc	hopper015
pcnc	up	7-00:00:00	1	idle	hopper016
pathogen	up	7-00:00:00	1	idle	hopper017
tc	up	7-00:00:00	3	mix	hopper[018-020]
tc	up	7-00:00:00	2	alloc	hopper[029-030]
tc	up	7-00:00:00	5	idle	hopper[021-025]
gold	up	7-00:00:00	2	idle	hopper[026-027]
fishgen	up	7-00:00:00	1	alloc	hopper028
neuro-hsc	up	7-00:00:00	6	alloc	hopper[031-036]
neuro-hsc	up	7-00:00:00	8	idle	hopper[037-044]
cup-ecs	up	7-00:00:00	2	alloc	hopper[049-050]
tid	up	7-00:00:00	1	alloc	hopper051
biocomp	up	7-00:00:00	1	alloc	hopper052
chakra	up	7-00:00:00	1	idle	hopper053
pna	up	7-00:00:00	1	down*	hopper045

```
[vanilla@hopper ~]$ srun --partition debug hostname
```



Tell slurm to run a program
on a compute node...

```
[vanilla@hopper ~]$ srun --partition debug hostname
```



Run the program on a
compute node in the
debug partition.

```
[vanilla@hopper ~]$ srun --partition debug hostname
```



The program
to run.

```
[vanilla@hopper ~]$ srun --partition debug hostname  
srun: Account not specified in script or  
~/.default_slurm_account, using latest project  
You have not been allocated GPUs. To request GPUs,  
use the -G option in your submission script.  
hopper011
```

```
[vanilla@hopper ~]$ queue
```

```
[vanilla@hopper ~]$ queue
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	ODELIST(REASON)
4314	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4315	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4317	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4318	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4319	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4320	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4321	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4322	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4323	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4324	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4325	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4326	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4328	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4329	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4330	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4331	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4332	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4333	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4334	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4335	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4336	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4337	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	

```
[vanilla@hopper ~]$ queue
```

JOBID	PARTITION	NAME	USER	ST	TIM
4314	general	PRE	erowland	PD	0:00
4315	general	PRE	erowland	PD	0:00
4317	general	PRE	erowland	PD	0:00
4318	general	PRE	erowland	PD	0:00

**PD means programs
that are waiting their
turn.**

Shows you what the slurm scheduler is doing right now.

Here we can see that user 'erowland' has a lot of programs waiting to run.

[illegible]


```
[vanilla@hopper ~]$ squeue -t R --all
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
4405	condo	2ndMA	vanilla	R	1-07:48:30	6	hopper[031-036]
5208	condo	NN	kgu	R	5:48:49	1	hopper015
5210	condo	NN	kgu	R	6:30:13	1	hopper014
5209	condo	NN	kgu	R	6:31:13	1	hopper013
5206	condo	NN	kgu	R	6:32:13	1	hopper051
5207	condo	NN	kgu	R	6:32:13	1	hopper052
5205	condo	NN	kgu	R	6:32:43	1	hopper028
4595	cup-ecs	golConfi	aalasand	R	2-06:51:59	1	hopper050
4594	cup-ecs	golConfi	aalasand	R	2-06:52:03	1	hopper049
5120	general	jupyterh	jacobm	R	11:45:47	1	hopper007
4313	general	PRE	erowland	R	1:17:29	2	hopper[003-004]
5111	general	1stMA	vanilla	R	11:15:28	2	hopper[005-006]
5025	general	c2n	jxzuo	R	1:50	1	hopper001
5024	general	c2n	jxzuo	R	31:28	1	hopper002
5203	general	NN	kgu	R	6:37:50	1	hopper009
5201	general	NN	kgu	R	6:38:14	1	hopper008
4390	tc	UCsTpCyd	lepluart	R	2-15:18:18	3	hopper[018-020]
5198	tc	NN	kgu	R	6:40:19	1	hopper030
5196	tc	NN	kgu	R	6:40:31	1	hopper029

```
[vanilla@hopper ~]$ srun --partition debug --ntasks 2 hostname  
srun: Account not specified in script or  
~/.default_slurm_account, using latest project  
You have not been allocated GPUs. To request GPUs, use the -G  
option in your submission script.  
hopper011  
hopper011
```

```
[vanilla@hopper ~]$ srun --partition debug --ntasks 2 hostname  
srun: Account not specified in script or  
~/.default_slurm_account, using latest project  
You have not been allocated GPUs. To request GPUs, use the -G  
option in your submission script.  
hopper011  
hopper011
```

You ran two **copies** of your program.

ntasks is the number of copies to run.

```
[vanilla@hopper ~]$ srun --partition debug --ntasks 2 --cpus-per-task 2 hostname  
srun: Account not specified in script or ~/.default_slurm_account, using latest project  
You have not been allocated GPUs. To request GPUs, use the -G option in your submission  
script.
```

```
hopper011
```

```
hopper011
```

**Here we are telling SLURM to
run 2 copies of our program
and let each copy of our
program use 2 CPUs.**

```
[vanilla@hopper ~]$ srun --partition debug --nodes 2 --ntasks-per-node 4 hostname  
srun: Account not specified in script or ~/.default_slurm_account, using  
latest project
```

```
hopper012
```

```
You have not been allocated GPUs. To request GPUs, use the -G option in  
your submission script.
```

```
hopper012
```

```
hopper011
```

```
hopper011
```

```
hopper012
```

```
hopper012
```

```
hopper011
```

```
hopper011
```

**Here we are telling SLURM to run 4
copies of our program on 2 different
compute nodes.**

**This is useful when our programs need
a bigger share of the compute node.**

```
[vanilla@hopper ~]$ srun --partition debug --nodes 2  
--ntasks-per-node 2 --cpus-per-task 2 hostname  
srun: Account not specified in script or  
~/.default_slurm_account, using latest project  
hopper011  
You have not been allocated GPUs. To request GPUs, use  
the -G option in your submission script.  
hopper011  
hopper012  
hopper012
```

And we can combine all three.

```
[vanilla@hopper ~]$ srun --partition debug --mem 4G  
--nodes 2 --ntasks-per-node 2 --cpus-per-task 2  
hostname
```

```
srun: Account not specified in script or  
~/default_slurm
```

```
hopper012
```

```
hopper012
```

```
You have not been granted access to  
the -G option in this partition
```

```
hopper011
```

```
Hopper011
```

**And we can specify how much
memory we want.**

**--mem 4G means give me 4
gigabytes of memory per node.**

```
[vanilla@hopper ~]$ srun --partition debug --mem 4G
--nodes 2 --ntasks-per-node 2 --cpus-per-task 2
hostname
srun: Account not specified in script or
~/default_slurm
hopper012
hopper012
You have not been granted access to the -G option in
hopper011
Hopper011
```

Why does all this matter?

The purpose of SLURM is to provide you the hardware your programs need.

So you have to understand what those requirements are really well.

```
[vanilla@hopper ~]$ srun --partition debug --mem 4G
--nodes 2 --ntasks-per-node 2 --cpus-per-task 2 hostname
srun: Account not specified in script or
~/default_slurm_account using latest project
hopper012
hopper012
You have not been assigned the
the -G option in the script
hopper011
Hopper011
```

- 1) Can my program use multiple CPUs?
- 2) How much memory does my program need?
- 3) Can my program use multiple compute nodes (MPI*, GNU Parallel*)?
- 4) Can my program use GPUs?

```
[vanilla@hopper ~]$ srun --partition debug --mem 4G  
--nodes 2 --ntasks-per-node 2 --cpus-per-task 2 hostname  
srun: Account not specified in script or  
~/ .default_slurm_account using latest project  
hopper012  
hopper012  
You have not been  
the -G option in  
hopper011  
Hopper011
```

This command is getting pretty long.

We can use **shell scripts** to automate
all this in **batch mode**.

Interactive vs Batch Mode

Interactive Mode

- Everything so far has been interactive. You request hardware, run your program, and get the output on your screen right away.

Batch Mode

- Most programs at an HPC center are run in “batch” mode.
- Batch mode means we write a shell script that the SLURM scheduler runs for us. The script requests hardware just like we did with `salloc` and then runs the commands in the script.
- Whatever would have been written to the screen is saved to a file instead.

```
[vanilla@hopper ~]$ git clone https://lobogit.unm.edu/CARC/workshops.git
Cloning into 'workshops'...
remote: Enumerating objects: 132, done.
remote: Counting objects: 100% (75/75), done.
remote: Compressing objects: 100% (43/43), done.
remote: Total 132 (delta 33), reused 74 (delta 32), pack-reused 57
Receiving objects: 100% (132/132), 57.58 KiB | 3.60 MiB/s, done.
Resolving deltas: 100% (51/51), done.
```

Rather than make you write shell scripts lets just download some we wrote for this workshop...

```
[vanilla@hopper ~]$ tree workshops
```

```
workshops/
├── intro_workshop
│   ├── code
│   │   ├── calcPiMPI.py
│   │   ├── calcPiSerial.py
│   │   └── vecadd
│   │       ├── Makefile
│   │       ├── vecadd_gpu.cu
│   │       ├── vecadd_mpi_cpu
│   │       ├── vecadd_mpi_cpu.c
│   │       ├── vecaddmpi_cpu.sh
│   │       └── vecadd_mpi_gpu.c
│   ├── data
│   │   ├── H20.gjf
│   │   └── step_sizes.txt
│   └── slurm
│       ├── calc_pi_array.sh
│       ├── calc_pi_mpi.sh
│       ├── calc_pi_parallel.sh
│       ├── calc_pi_serial.sh
│       ├── gaussian.sh
│       ├── hostname_mpi.sh
│       ├── vecadd_hopper.sh
│       ├── vecadd_xena.sh
│       ├── workshop_example2.sh
│       ├── workshop_example3.sh
│       └── workshop_example.sh
└── README.md
```

Run tree to see how the workshops directories are organized...

```
[vanilla@hopper ~]$ tree workshops
```

```
workshops/
├── intro_workshop
│   ├── code
│   │   ├── calcPiMPI.py
│   │   ├── calcPiSerial.py
│   │   └── vecadd
│   │       ├── Makefile
│   │       ├── vecadd_gpu.cu
│   │       ├── vecadd_mpi_cpu
│   │       ├── vecadd_mpi_cpu.c
│   │       ├── vecaddmpi_cpu.sh
│   │       └── vecadd_mpi_gpu.c
│   ├── data
│   │   ├── H20.gjf
│   │   └── step_sizes.txt
│   └── slurm
│       ├── calc_pi_array.sh
│       ├── calc_pi_mpi.sh
│       ├── calc_pi_parallel.sh
│       ├── calc_pi_serial.sh
│       ├── gaussian.sh
│       ├── hostname_mpi.sh
│       ├── vecadd_hopper.sh
│       ├── vecadd_xena.sh
│       ├── workshop_example2.sh
│       ├── workshop_example3.sh
│       └── workshop_example.sh
└── README.md
```

Run **tree** to see how the workshops directories are organized...

The workshop files are divided into “code”, “slurm”, and “data” directories.

You will also see a **radio_astronomy** directory

```
[vanilla@hopper intro_workshop]$ pwd
/users/vanilla/workshops/intro_workshop
[vanilla@hopper intro_workshop]$ cat slurm/workshop_example.sh
#!/bin/bash
#SBATCH --partition debug
#SBATCH --ntasks 4
#SBATCH --time 00:05:00
#SBATCH --job-name ws_example
#SBATCH --mail-user your_username@unm.edu
#SBATCH --mail-type ALL
```

```
hostname
```

Let's take a look at the **workshop_example.sh** script in the slurm directory...

```
[vanilla@hopper intro_workshop]$ sbatch slurm/workshop_example.sh
sbatch: Account not specified in script or ~/.default_slurm_account,
using latest project
Submitted batch job 5252
[vanilla@hopper intro_workshop]$
```

We **submit** our slurm
shell script with the
sbatch command.

```
[vanilla@hopper intro_workshop]$ sbatch slurm/workshop_example.sh
sbatch: Account not specified in script or ~/.default_slurm_account,
using latest project
Submitted batch job 5252
[vanilla@hopper intro_workshop]$
```

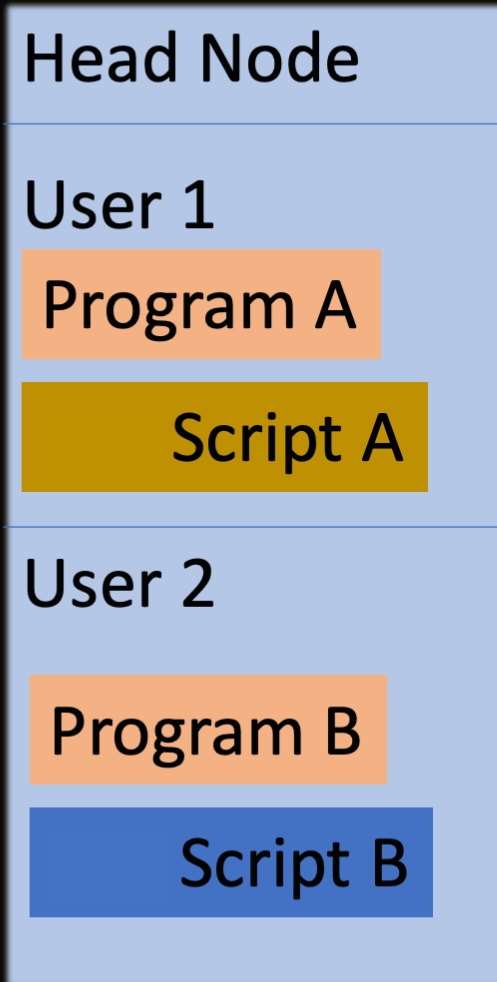
Notice that the only output we get is a job id.

This indicates that the script was successfully sent to the scheduler.

The commands in the script will run as soon as the hardware requested is available.

We **submit** our slurm shell script with the sbatch command.

Workflow



Compute Node 01

Compute Node 02

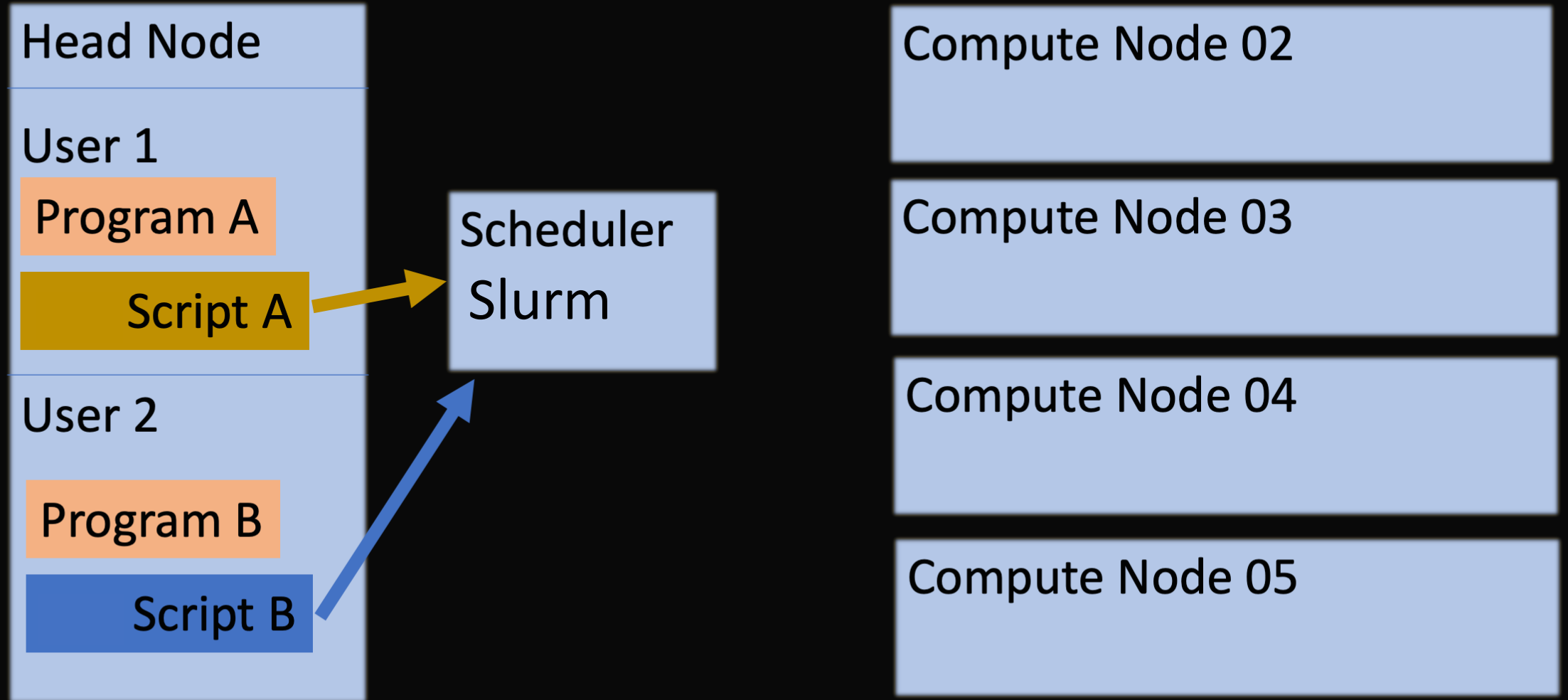
Compute Node 03

Compute Node 04

Compute Node 05

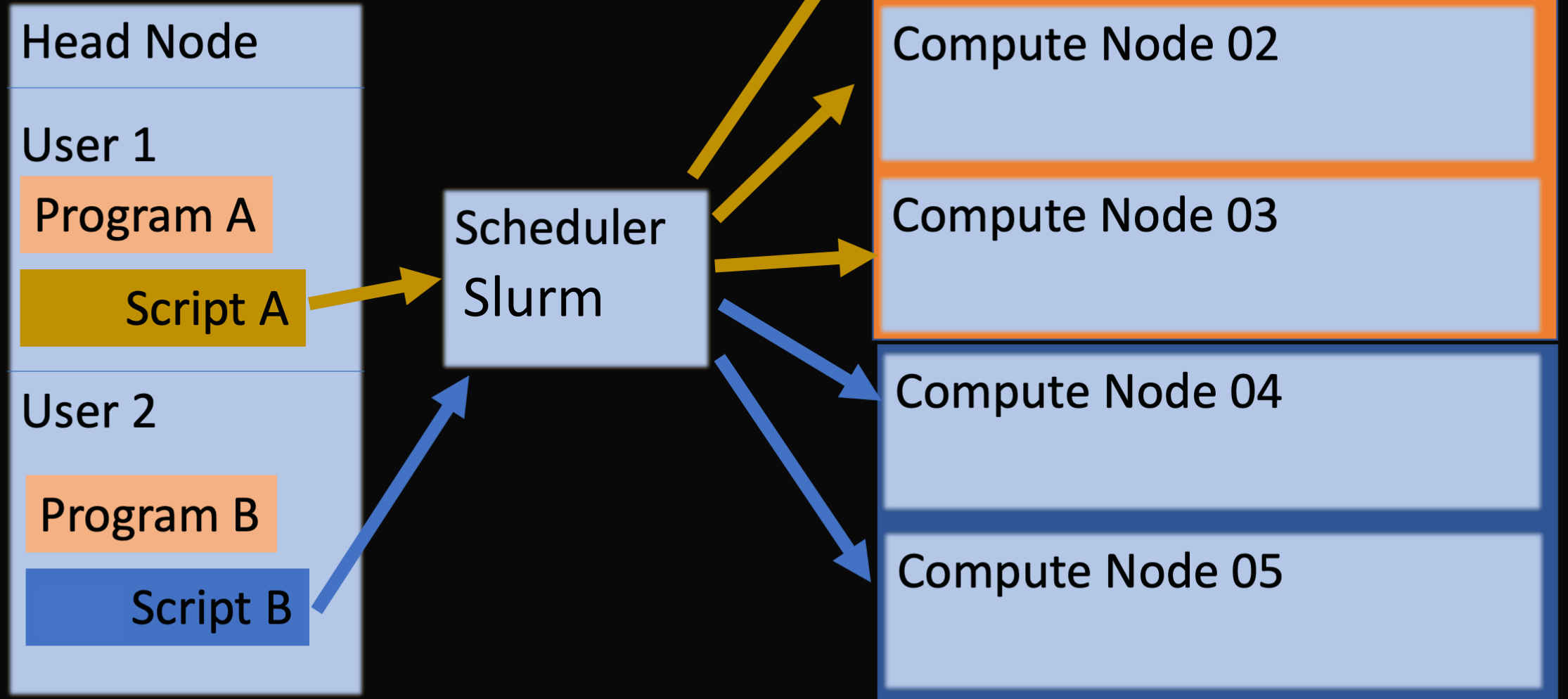
Shared filesystems – All nodes can access the same programs and write output

Workflow



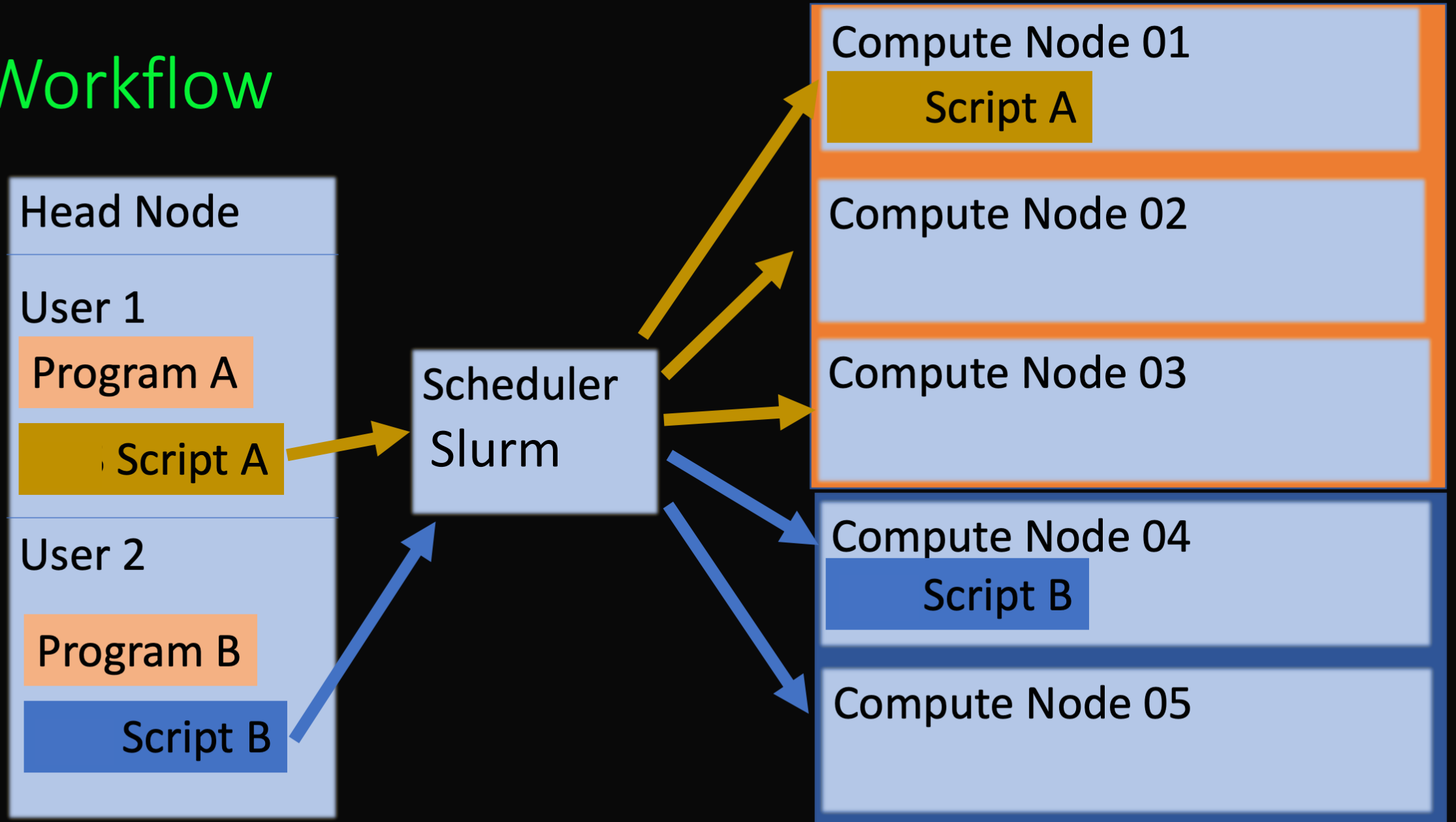
Shared filesystems – All nodes can access the same programs and write output

Workflow



Shared filesystems – All nodes can access the same programs and write output

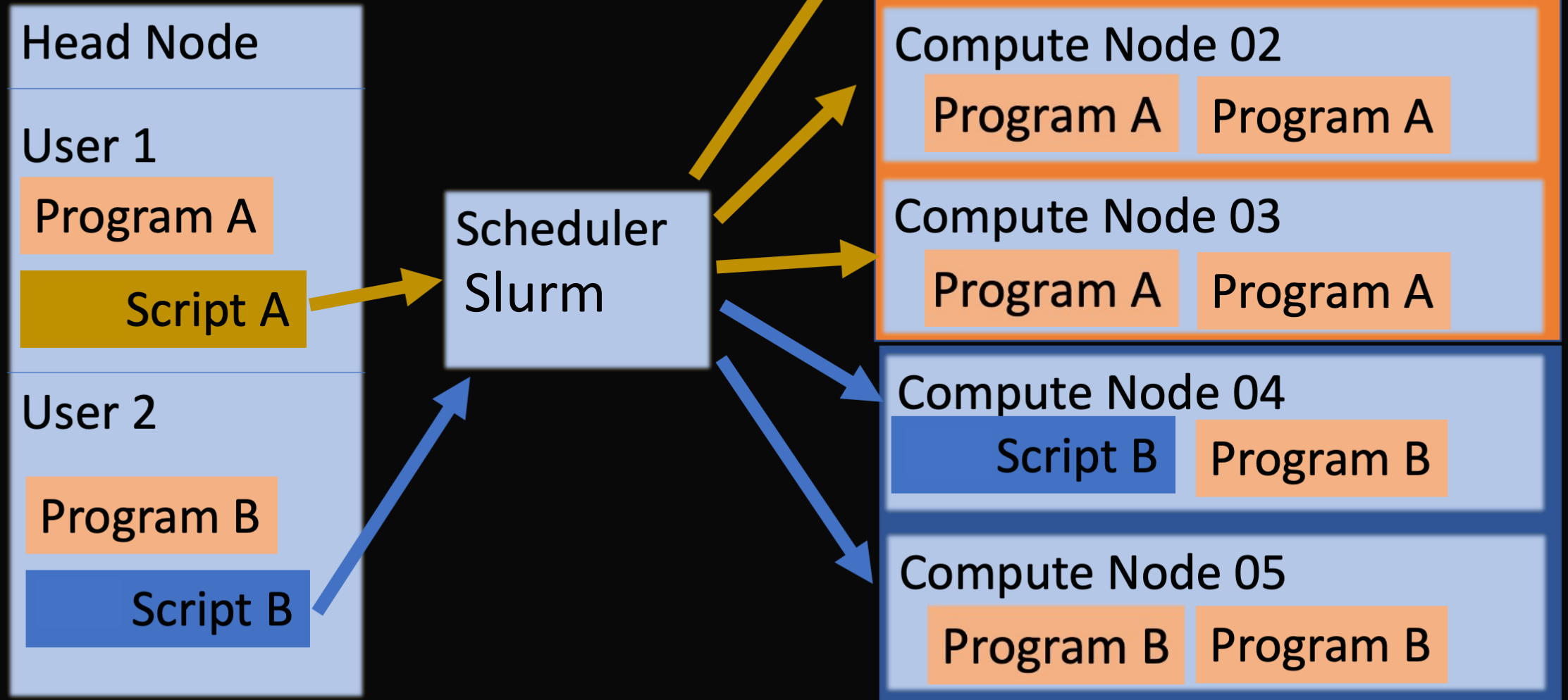
Workflow



Shared filesystems – All nodes can access the same programs and write output

Workflow

We need something in the script to run the program on all the nodes. E.g. srun.



Shared filesystems – All nodes can access the same programs and write output

```
[vanilla@hopper intro_workshop]$ ls  
code  data  pbs   slurm  slurm-5252.out
```

The **hostname** command is very fast so everyone's job should finish in a few seconds.

When it is finished you will have a new file named `slurm-{your job id}.out`.



```
[vanilla@hopper intro_workshop]$ ls  
code  data  pbs  slurm  slurm-5252.out
```



When it is finished you will have a new file named slurm-{your job id}.out.

```
[vanilla@hopper intro_workshop]$ cat slurm-5252.out  
hopper011
```

```
[vanilla@hopper intro_workshop]$ ls  
code  data  pbs  slurm  slurm-5252.out
```



When it is finished you will have a new file named `slurm-{your job id}.out`.

```
[vanilla@hopper intro_workshop]$ cat slurm-5252.out  
hopper011
```

Why did it only run the program once instead of 4 times?

```
[vanilla@hopper intro_workshop]$ sbatch slurm/workshop_example1.sh  
sbatch: Account not specified in script or ~/.default_slurm_account,  
using latest project  
Submitted batch job 5252  
[vanilla@hopper intro_workshop]$
```

**Take a look at the
output file.**

**You should see 4
hostname output lines.**



Python based Common Astronomy Software Applications

```

matthew — IPython: users/mfricke — ssh hopper — 80x24
~ — zsh
Last login: Sun Feb 19 15:12:04 2023 from 50.193.238.94
mfricke@hopper:~ $ srun --partition debug --pty bash
srun: Using account 2016199 from ~/.default_slurm_account
You have not been allocated GPUs. To request GPUs, use the -G option in your sub
mission script.
Job 1817180 running on hopper011
mfricke@hopper011:~ $ module load casa
mfricke@hopper011:~ $ casa --nogui

optional configuration file config.py not found, continuing CASA startup without
it

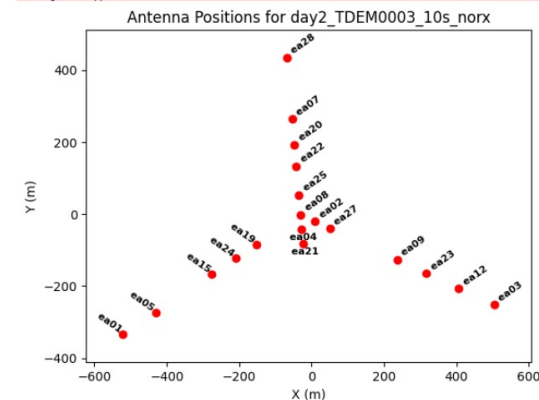
IPython 7.15.0 -- An enhanced Interactive Python.

Using matplotlib backend: agg
Creating a new telemetry file
Telemetry initialized. Telemetry will send anonymized usage statistics to NRAO.
You can disable telemetry by adding the following line to the config.py file in
your rcdir (e.g. ~/.casa/config.py):
telemetry_enabled = False
--> CrashReporter initialized.
CASA 6.5.2.26 -- Common Astronomy Software Applications [6.5.2.26]

CASA <1>: █

```

/users/mfricke/.local/lib/python3.8/site-packages/casatasks/private/task_plotants.py:356: UserWarning: Matplotlib is currently using module://m
atplotlib_inline.backend_inline, which is a non-GUI backend, so cannot show the figure.
fig.show()



```

In [8]: flagdata(vis=observation_path,
mode='list',
infile=["field='2,3' antenna='ea12' timerange='03:41:00-04:10:00'",
"field='2,3' antenna='ea07,ea08' timerange='03:21:40-04:10:00' spw='1'"])

```

Out[8]: {}

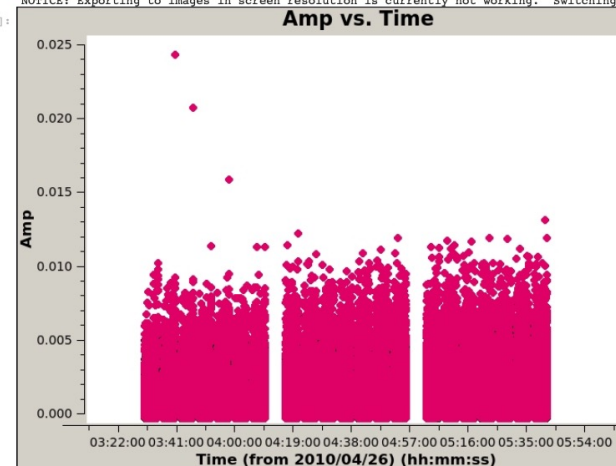
```

In [14]: plotms(vis=observation_path,field='3',
plotfile='templ.jpg',
xaxis='time',yaxis='amp',correlation='RR,LL',
avgchannel='64',spw='0-1:4-60', coloraxis='spw')
c

```

2023-02-19 01:09:00 WARN MStTransformManager::initDataSelectionParams Number of selected channels 57 for SPW 0 is smaller than specif
ied chanbin 64
2023-02-19 01:09:00 WARN MStTransformManager::initDataSelectionParams+ Setting chanbin to 57 for SPW 0
2023-02-19 01:09:00 WARN MStTransformManager::initDataSelectionParams+ Number of selected channels 57 for SPW 1 is smaller than specif
ied chanbin 64
2023-02-19 01:09:00 WARN MStTransformManager::initDataSelectionParams+ Setting chanbin to 57 for SPW 1
NOTICE: Exporting to images in screen resolution is currently not working. Switching to high resolution (which is slower, but works).

Out[14]:



```

In [15]: plotms(vis=observation_path,field='3',plotfile='temp2.jpg',
xaxis='uvdist',yaxis='amp',correlation='RR,LL',
avgchannel='64',spw='0-1:4-60', coloraxis='spw')
Image(filename='temp2.jpg')

```

```
[vanilla@hopper ~]$ cd ~/workshops/radio_astronomy/  
[vanilla@hopper radio_astronomy]$
```

```
$rsync -a --info=progress2 /projects/shared/workshops/radio_astronomy/casa/* .  
3,137,159,512 100% 169.43MB/s 0:00:17 (xfr#233, to-chk=0/271)
```

**Copy example
observation to your
home storage ...**



```
[vanilla@hopper radio_astronomy]$ srun --partition debug --pty bash
srun: Account not specified in script or ~/.default_slurm_account, using latest project
You have not been allocated GPUs. To request GPUs, use the -G option in your submission
script.
```

```
Job 1817350 running on hopper011
```

```
[vanilla@hopper011 radio_astronomy]$ module load casa
```

```
[vanilla@hopper011 radio_astronomy]$ casa --nogui
```

```
optional configuration file config.py not found, continuing CASA startup without it
```

```
IPython 7.15.0 -- An enhanced Interactive Python.
```

```
Using matplotlib backend: <object object at 0x1502a77a2590>
```

```
Telemetry initialized. Telemetry will send anonymized usage statistics to NRAO.
```

```
You can disable telemetry by adding the following line to the config.py file in your
rcdir (e.g. ~/.casa/config.py):
```

```
telemetry_enabled = False
```

```
--> CrashReporter initialized.
```

```
CASA 6.5.2.26 -- Common Astronomy Software Applications [6.5.2.26]
```

```
CASA <1>:
```

CASA <1>: default listobs

CASA <2>: inp

```
# listobs -- List the summary of a data set in the logger or in a file
vis          = ''          # Name of input visibility file (MS)
selectdata   = True        # Data selection parameters
  spw         = ''         # Selection based on spectral-window/frequency/channel.
  field       = ''         # Selection based on field names or field index numbers. Default is all.
  antenna     = ''         # Selection based on antenna/baselines. Default is all.
  uvrange     = ''         # Selection based on uv range. Default: entire range. Default units: meters.
  timerange   = ''         # Selection based on time range. Default is entire range.
  correlation  = ''         # Selection based on correlation. Default is all.
  scan        = ''         # Selection based on scan numbers. Default is all.
  intent      = ''         # Selection based on observation intent. Default is all.
  feed        = ''         # Selection based on multi-feed numbers: Not yet implemented
  array       = ''         # Selection based on (sub)array numbers. Default is all.
  observation  = ''         # Selection based on observation ID. Default is all.
verbose      = True        # Controls level of information detail reported. True reports more than False.
listfile     = ''          # Name of disk file to write output. Default is none (output is written to
                           # logger only).
listunfl     = False       # List unflagged row counts? If true, it can have significant negative
                           # performance impact.
cachesize    = 50.0        # EXPERIMENTAL. Maximum size in megabytes of cache in which data structures can
                           # be held.
```

CASA <5>: vis='day2_TDEM0003_10s_norx'

CASA <6>: go

Out[6]:

```
{'BeginTime': 55312.140173611115,  
 'EndTime': 55312.256215277775,  
 'IntegrationTime': 10026.0,  
 'field_2': {'code': 'D',  
   'direction': {'m0': {'unit': 'rad', 'value': 2.5959462699},  
     'm1': {'unit': 'rad', 'value': 0.3093655364000001},  
     'refer': 'J2000',  
     'type': 'direction'},  
   'name': 'J0954+1743'},  
 'field_3': {'code': 'NONE',  
   'direction': {'m0': {'unit': 'rad', 'value': 2.5654436141},  
     'm1': {'unit': 'rad', 'value': 0.2317441393},  
     'refer': 'J2000',  
     'type': 'direction'},  
   'name': 'IRC+10216'},
```

```
CASA <9>: default plotants
```

```
CASA <10>: vis='day2_TDEM0003_10s_norx'
```

```
CASA <11>: figfile='ant_locations.png'
```

```
CASA <12>: inp
```

```
# plotants -- Plot the antenna distribution in the local reference frame:
```

```
vis                = 'day2_TDEM0003_10s_norx'
```

```
# Name of input visibility file (MS)
```

```
figfile            = 'ant_locations.png'
```

```
# Save the plotted figure to this file
```

```
antindex           = False
```

```
# Label antennas with name and antenna ID
```

```
logpos             = False
```

```
# Whether to plot logarithmic positions
```

```
exclude            = ''
```

```
# Antenna name/id selection to exclude from
```

```
plot
```

```
checkbaselines     = False
```

```
# Whether to check baselines in the main
```

```
table.
```

```
title              = ''
```

```
# Title for the plot
```

```
showgui            = True
```

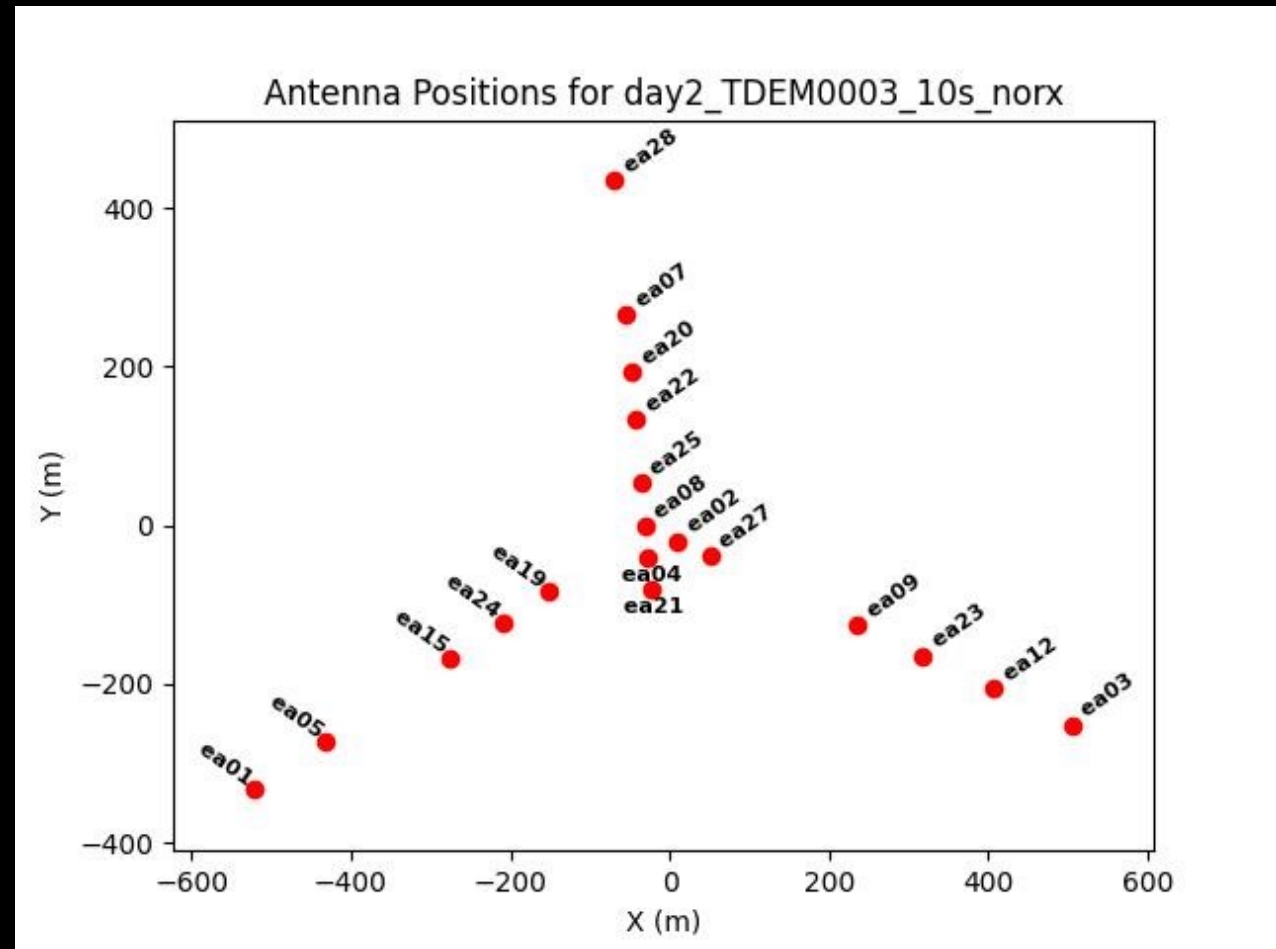
```
# Show plot on gui.
```

```
CASA <13>: go
```

```
CASA <14>: quit()
```

```
[vanilla@hopper011 radio_astronomy]$ exit
```

```
[vanilla@hopper radio_astronomy]$ file ant_locations.png
ant_locations.png: PNG image data, 640 x 480, 8-bit/color RGBA, non-interlaced
```



```
[vanilla@hopper radio_astronomy]$ cat casa_example1.sh  
#!/bin/bash
```

```
#SBATCH --partition debug  
#SBATCH --nodes 1  
#SBATCH --ntasks-per-node 4  
#SBATCH --time 00:10:00  
#SBATCH --job-name CASAEX1  
#SBATCH --mail-user your_username@unm.edu  
#SBATCH --mail-type ALL
```

```
module load lsb_release  
module load casa
```

```
mpicasa -n $SLURM_NTASKS xvfb-run -d casa -c casa_example1.py
```

```
[vanilla@hopper radio_astronomy]$ cat casa_example1.py
visfile = 'day2_TDEM0003_10s_norx'

listobs(vis = visfile, listfile = 'casa_example1.list')

plotants(vis=visfile, figfile='example1_ant_locations.jpg')

plotms(vis=visfile,
        xaxis='time',yaxis='elevation',correlation='RR,LL',
        plotfile='time_elevation.jpg',
        avgchannel='64',spw='0:4~60', coloraxis='field')

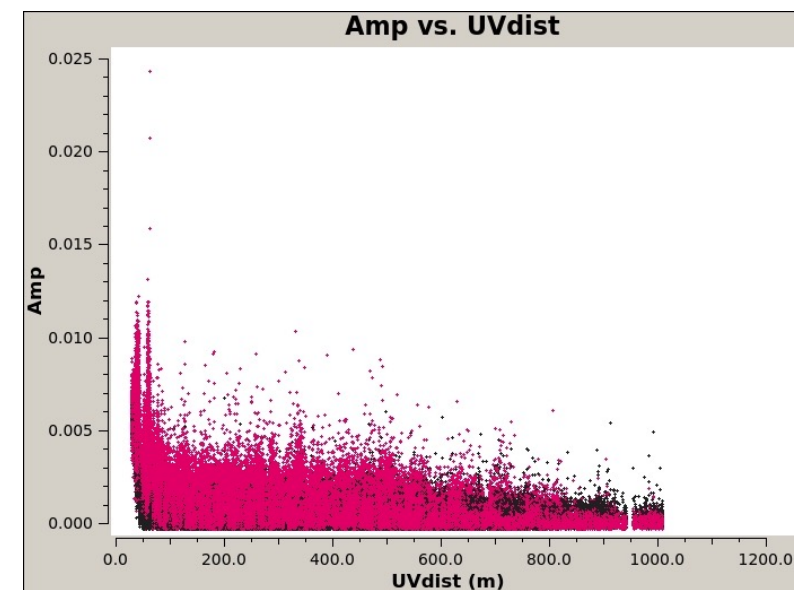
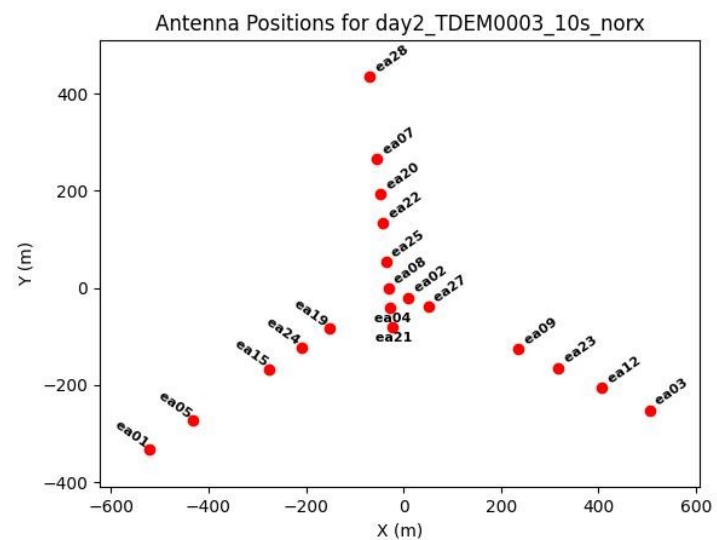
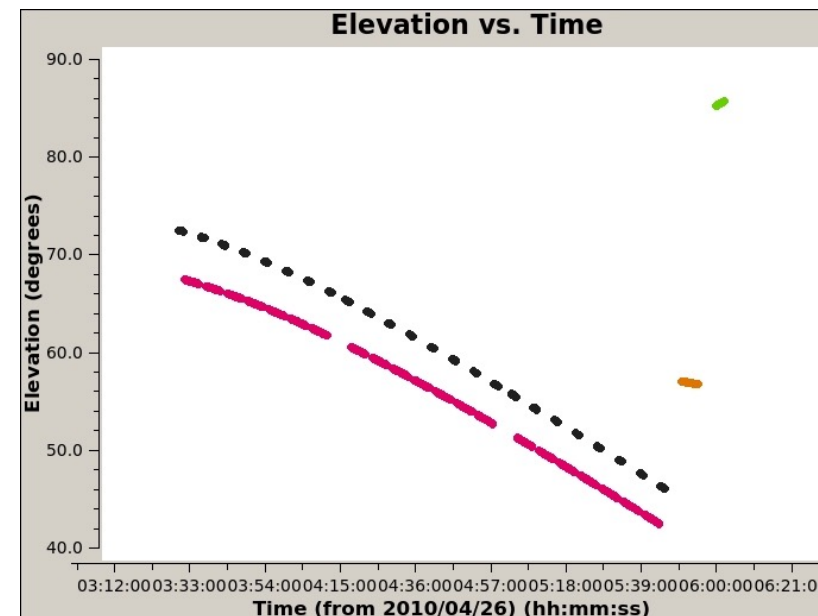
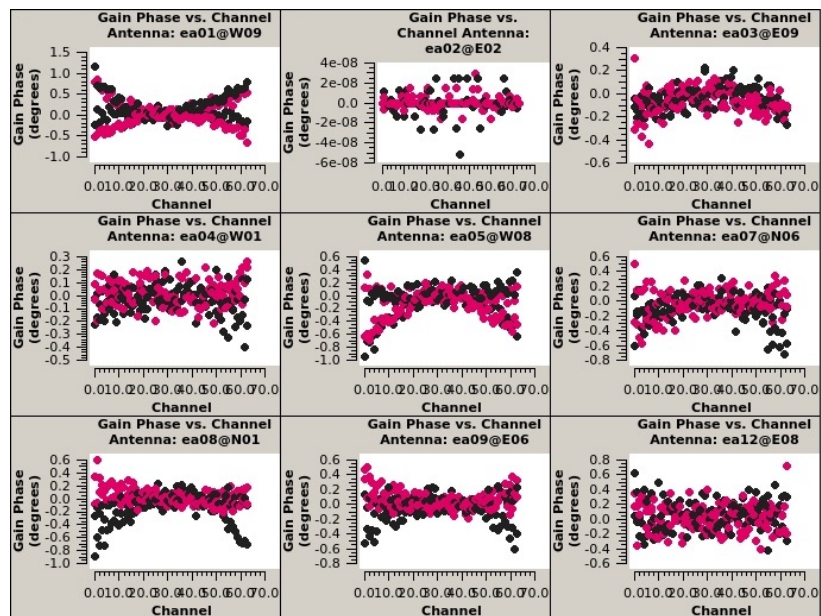
plotms(vis=visfile,
        xaxis='time',yaxis='amp',correlation='RR,LL',
        plotfile='time_amp.jpg',
        avgchannel='64',spw='0:4~60', coloraxis='field')
```

<snip>

```
[vanilla@hopper radio_astronomy]$ sbatch casa_example1.sh
```

```
[vanilla@hopper radio_astronomy]$ squeue --me
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
1817355	debug	CASAEX1	vanilla	R	0:14	1	hopper011



```
[vanilla@hopper radio_astronomy]$ module load miniconda3
```

```
[vanilla@hopper radio_astronomy]$ conda env create -f  
casa_env.yml
```

Install modular casa



```
[vanilla@hopper radio_astronomy]$ source activate casa

(casa) [vanilla@hopper radio_astronomy]$ python -m casatools --update-user-data
attempting to install/update runtime data in /users/vanilla/.casa/data
.....
.....
```

Install casa user data...



```
(casa) [vanilla@hopper radio_astronomy]$ pip install  
pyvirtualdisplay
```

```
(casa) [vanilla@hopper radio_astronomy]$
```

Install modular casa





<https://hopper.alliance.unm.edu:8000>





https://hopper.alliance.unm.edu:8000



Sign in

Username:

Password:

Sign in



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Home

Token

Admin

mfricke

Logout

Server Options

Select a job profile:

Debug Queue, 1 hours, 1 core, 4GB RAM



Start

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Control Panel

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☐	 workshops	6 hours ago	
☐	 workshops3	7 months ago	
☐	 workshops_prev	5 months ago	
☐	 wrapspawner	3 years ago	
☐	 xena_ganglia	2 years ago	
☐	 xena_ganglia_backup_20200842	3 years ago	
☐	 xena_matlab_modules	a year ago	
☐	 yolov5	4 months ago	
☐	 Capstone Project.ipynb	2 years ago	213 kB



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Select items to perform actions on them.

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36 minutes ago



README.md

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47 B





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Logout

Control Panel

Files

Running

Clusters

Select items to perform actions on them.

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☐ 0 ▾	📁 / workshops / radio_astronomy	Name ▾	Last Modified	File size
	📁 ..		seconds ago	
☐	📁 sis14_twhya_calibrated_flagged.ms		6 years ago	
☐	📁 temp		an hour ago	
☐	📄 casa_example2.ipynb		2 hours ago	107 kB
☐	📄 casa_env.yml		38 minutes ago	2.7 kB
☐	📄 casa_example1.py		an hour ago	3.75 kB
☐	📄 casa_example1.py.bad		an hour ago	13.5 kB
☐	📄 casa_example1.sh		2 hours ago	307 B

workshops/radio_astronomy/ casa_example2 - Jupyter Notebook

Trusted Python [conda env:.conda-casa]

File Edit View Insert Cell Kernel Widgets Help

Interrupt Restart Restart & Clear Output Restart & Run All Reconnect Shutdown Change kernel

CASA needs a display

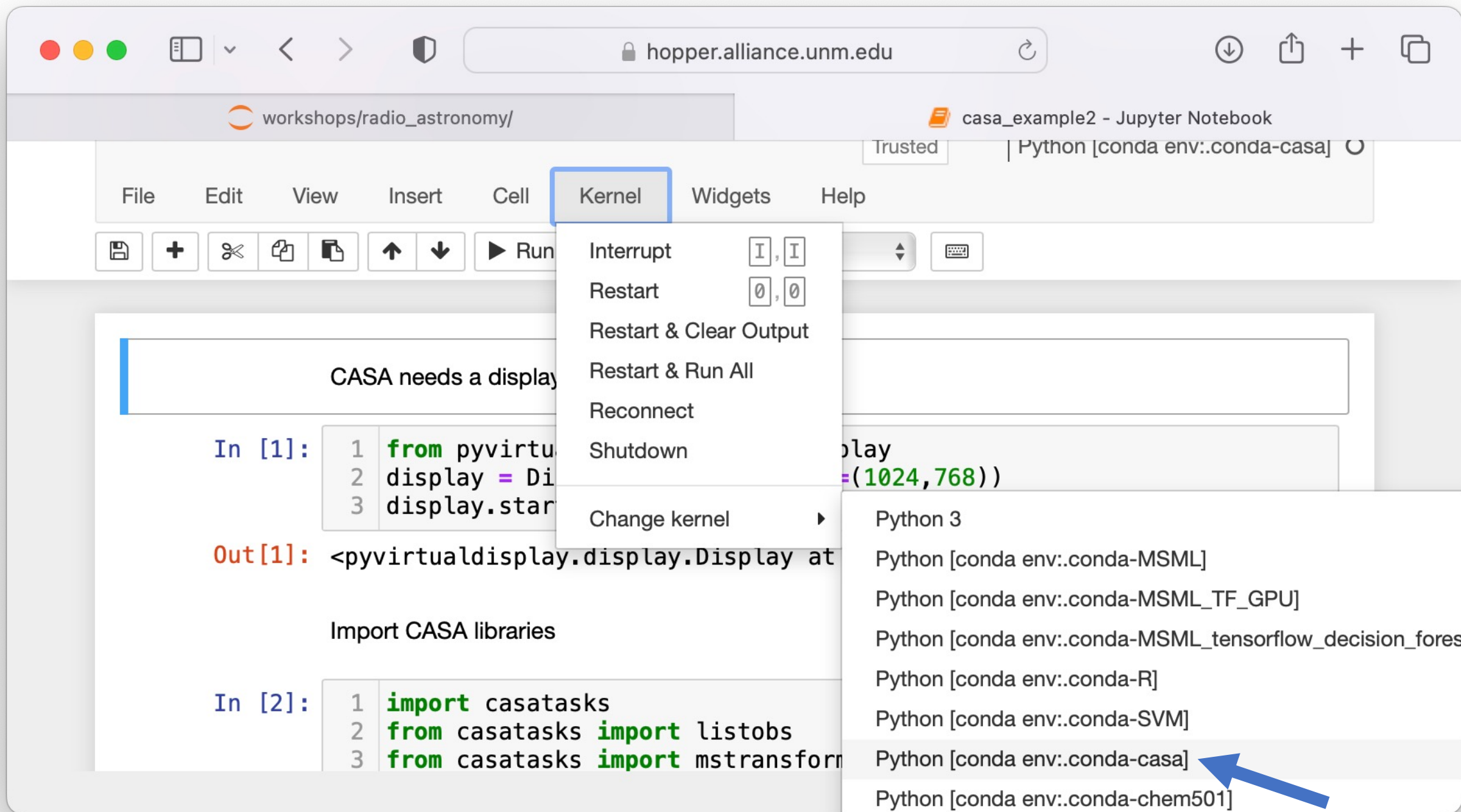
```
In [1]: 1 from pyvirtu
        2 display = Di
        3 display.star
```

```
Out[1]: <pyvirtualdisplay.display.Display at
```

Import CASA libraries

```
In [2]: 1 import casatasks
        2 from casatasks import listobs
        3 from casatasks import mstransform
```

Python 3
Python [conda env:.conda-MSML]
Python [conda env:.conda-MSML_TF_GPU]
Python [conda env:.conda-MSML_tensorflow_decision_forest]
Python [conda env:.conda-R]
Python [conda env:.conda-SVM]
Python [conda env:.conda-casa]
Python [conda env:.conda-chem501]



workshops/radio_astronomy/ casa_example2 - Jupyter Notebook

Trusted Python [conda env:.conda-casa]

File Edit View Insert Cell Kernel Widgets Help

Run Code

CASA needs a display

```
In [ ]: 1 from pyvirtualdisplay import Display
        2 display = Display(visible=0, size=(1024, 768))
        3 display.start( )
```

Import CASA libraries

```
In [ ]: 1 import casatasks
        2 from casatasks import listobs
        3 from casatasks import mstransform
        4 from casatasks import tclean
        5 from casaviewer import imview
```

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workshops/radio_astronomy/

casa_example2 - Jupyter Notebook

Restart kernel and re-run the whole notebook?

Are you sure you want to restart the current kernel and re-execute the whole notebook? All variables and outputs will be lost.

Continue Running

Restart and Run All Cells

File

from pyvirtualdisplay import Display

2 display = Display(visible=0, size=(1024, 768))

3 display.start()

Import CASA libraries

In []:

1 import casatasks

2 from casatasks import listobs

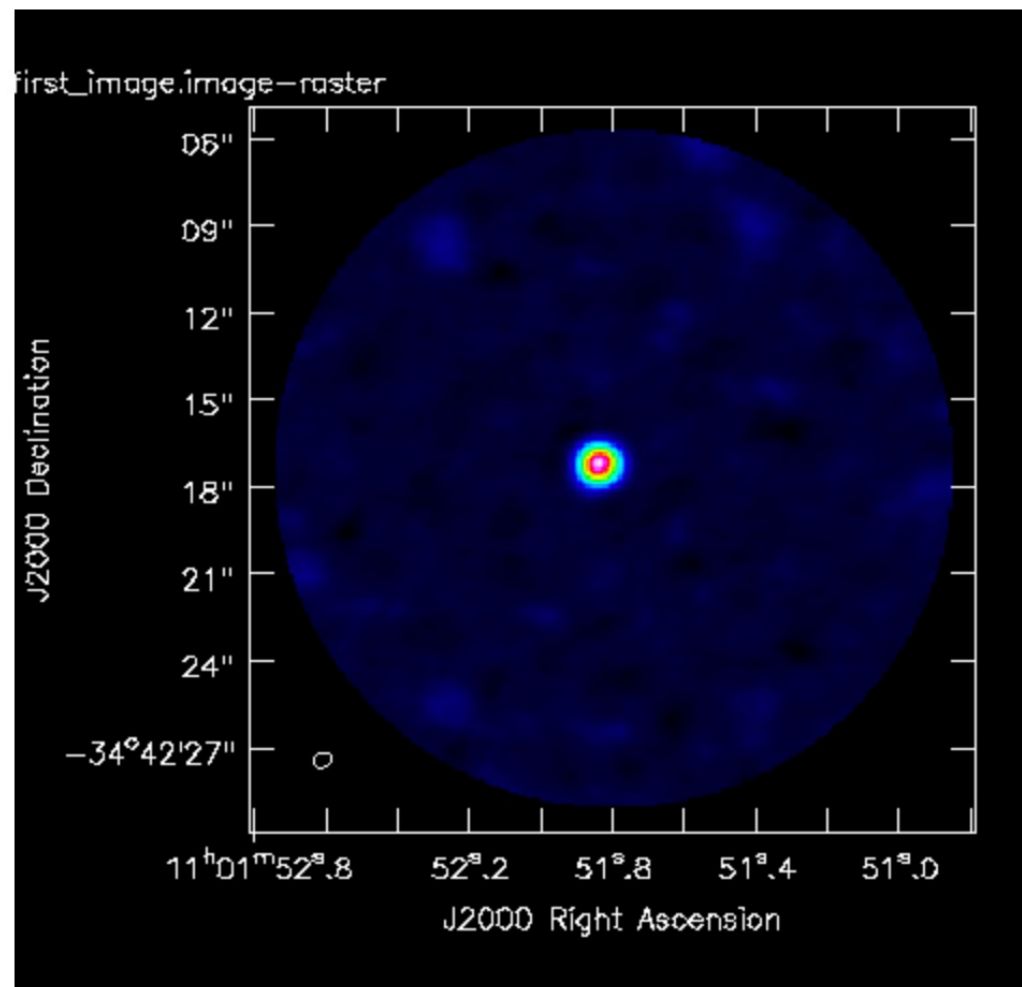
3 from casatasks import mstransform

4 from casatasks import tclean

5 from casaviewer import imview


```
In [10]: 1 Image(filename="test.png")
```

```
Out[10]:
```



You have learned

- how to run programs using the SLURM scheduler
- the difference between interactive and batch jobs
- how to check the status of your jobs
- how to ask for the hardware resources you need
- you ran CASA interactively in the terminal and in jupyterhub and in batch mode

